11.13 The Snowplough Problem

WORKSHEET

The question

STEP III Specimen paper, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q11

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself The first plough meets snow of depth $h + kt$ at time t. What is its speed? Is it easier to find x as a function of t, or t as a function of x?

STAGE 2 The second plough's equation

Ask yourself The second plough reaches position y at time t. How deep is the snow there? (What was the depth just after the first plough passed?)

STAGE 3 Solving

Ask yourself This is linear in t: $\frac{dt}{dy} - \alpha t = -\frac{\alpha h}{k}(e^{\alpha y} - 1)$. What integrating factor works? What is the starting condition?

STAGE 4 The collision

Ask yourself The ploughs collide when both reach the same point at the same time. Compare $t_2(y)$ with $t_1(y)$.

Try this next

1. **Agnew's problem.** Snow starts falling at a steady rate at an unknown time before noon. A snowplough whose speed is inversely proportional to the depth starts at noon and goes 2 miles in the first hour and 1 mile in the second. Show that if it started snowing t_0 hours before noon, then $(1 + \frac{2}{t_0})^2 = (1 + \frac{1}{t_0})^3$, deduce that $\frac{1}{t_0}$ is the golden ratio, and find when it started snowing.

2. **Verification.** Verify by differentiation that $t_2(y)$ satisfies the differential equation of part (ii).

3. **The time gap.** Show that if $kT > h$, the time gap $t_2(y) - t_1(y)$ first increases and then decreases to zero, and find where it is greatest. What happens if $kT \leq h$? Interpret this.

4. **Three ploughs.** A third plough starts at time $2T$. Write down the equation for its motion. Does it catch the second plough before or after the second catches the first?

Further reading

- **Ralph Palmer Agnew, *Differential Equations*** (McGraw-Hill, 1942). The original snowplough problem.
- **George Simmons, *Differential Equations with Applications and Historical Notes*** (CRC Press), which discusses the snowplough problem among many others.
- **Richard Haberman, *Mathematical Models*** (SIAM). Traffic flow and characteristics, at an accessible level.

The full guided solution, and what the question leads to at university, is in Section 11.13:
https://beyondstep.keithmcnulty.org/ch11/snowplough.html